IoT-based Behavioral Health Monitoring and Support System

PROJECT REPORT

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IoT-based Behavioral Health Monitoring and Support System

A Project Report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

In

Electronics and Instrumentation Engineering

By

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Certificate

This is to certify that the Project entitled "***IoT- based behavioral health monitoring and support system***" has been carried out by ***Ankita Das*** (23beih27), ***Nupur Das*** (23beia96), ***Trupti Dash***(23beia58) and ***Abhijit Mohanty*** (23beid03) under my guidance and supervision and be accepted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Electronics and Instrumentation Engineering** under Silicon University, Bhubaneswar.

Mr. Debi Prasad Pradhan (Designation)	Dr. Nalini Singh Project Coordinator	Dr. Sudhanshu Pati H.O.D, Dept. of EE

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been include. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/sources in our submission. We understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Finally I would like to thank all who's direct and indirect support helped me to complete our project work.

Ankita Das

Nupur Das

Trupti Dash

Abhijit Mohanty

Department Vision, Mission and PEOs

Vision…

To be recognized as a beacon of quality education and research in the field of Electronics and Instrumentation Engineering.

Mission…

1. Continually improve the standard of our graduates by having high caliber motivated faculty members together with quality educational programs and facilities in-line with the rapid technological advancements in the field of Electronics and Instrumentation Engineering (Knowledge, Skill and Quality).

2. Provide a balanced regime of quality education that incorporates theoretical and practical education, innovation and creativity as well as freedom of thought and research with emphasis on professionalism and ethical behavior (Professionalism & Ethics).

3. Promote and support research activities over a broad range of academic interests among students and staff for growth of individual knowledge and prepares for continuous learning (Research and Life-long Learning).

PEOs of the Program

PEO1. Fundamental Knowledge & Core Competence: To provide knowledge of science and engineering fundamental for an electronics & communication engineer and equip with proficiency of mathematical foundations and inculcate competent problem solving ability.

PEO2. Competency for Real World: To design, optimize and maintain electronics and instrumentation systems in tune with community needs and environmental concerns.

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PEO4. Life-long Learning: To pursue higher studies or engage in a technical or managerial role in diverse teams, grow professionally in their career through continued education & training of technical and management skills.

Abstract

This project presents an IoT-based Behavioral Health Monitoring and Support System designed to enable continuous, data-driven, and personalized mental health care. In modern society, issues such as stress, anxiety, and depression are increasingly prevalent but often remain undiagnosed due to the limitations of traditional assessment methods.

The proposed system utilizes IoT and AI technologies to monitor key physiological and behavioral indicators, transmitting the data to a cloud platform for intelligent analysis. Machine learning algorithms interpret subtle variations in emotional states, providing real-time alerts, feedback, and personalized coping strategies through a mobile application. A professional dashboard further supports healthcare practitioners with continuous insights for early intervention.

By combining smart sensing, cloud analytics, and adaptive feedback, this system shifts mental healthcare from reactive treatment to proactive management. It offers a non-invasive, affordable, and accessible approach suitable for personal wellness, remote patient monitoring, and clinical applications, fostering a more connected and compassionate digital health ecosystem.

Keywords- *Place 4 to 5 Keywords.*

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Introduction

Chapter 1

Introduction

1.1 Introduction to IoT-Based Behavioral Health Monitoring and Support System

The Internet of Things (IoT) has emerged as one of the most significant technological advancements of the 21st century, connecting billions of devices capable of sensing, processing, and transmitting data. In recent years, IoT has found tremendous applications in healthcare, enabling continuous patient monitoring, remote diagnostics, and data-driven medical decision-making. One of the most promising applications of IoT in healthcare lies in **behavioral and mental health monitoring**, where continuous tracking of physiological signals can reveal valuable insights into a person’s emotional and psychological state.

Behavioral health includes mental, emotional, and social well-being, influencing how individuals think, feel, and act. Common disorders such as anxiety, depression, and chronic stress affect millions of people worldwide, often going undetected due to the lack of continuous monitoring tools. Traditional diagnostic methods rely on patient self-assessment and clinical interviews, which are subjective and infrequent. Consequently, many early warning signs remain unnoticed until conditions worsen.

An IoT-enabled behavioral health monitoring system aims to overcome these limitations by integrating **wearable sensors, wireless communication, cloud computing, and data analytics**. This combination allows real-time acquisition and analysis of physiological parameters such as heart rate variability (HRV), galvanic skin response (GSR) and activity levels each of which correlates with an individual’s mental and emotional status. These parameters are processed using intelligent algorithms to detect stress, anxiety, and other behavioral changes.

The system architecture consists of five layers:

1. **Sensing Layer** – Wearable sensors measure physiological and behavioral data such as heart rate, SpO_2, electrodermal activity (EDA) and body movement.
2. **Edge Layer** – The ESP32 microcontroller acts as the edge device, performing initial signal processing, noise filtering, and temporary data storage before transmission.
3. **Communication Layer** – Manages wireless data transfer between the edge device and the cloud using Wi-Fi or Bluetooth protocols, ensuring secure and low-latency communication.
4. **Cloud Layer** – Provides storage, processing, and analytics of sensor data using algorithms and machine learning models to identify behavioral patterns such as stress or anxiety.
5. **Application Layer** – Data analytics and visualization are performed to provide real-time feedback, insights, and alerts to users and healthcare professionals.

The interaction between these layers enables a continuous and intelligent behavioral health monitoring process. Data collected from sensors is preprocessed at the edge, transmitted securely through the communication layer, analyzed in the cloud, and finally visualized for end users through intuitive applications.

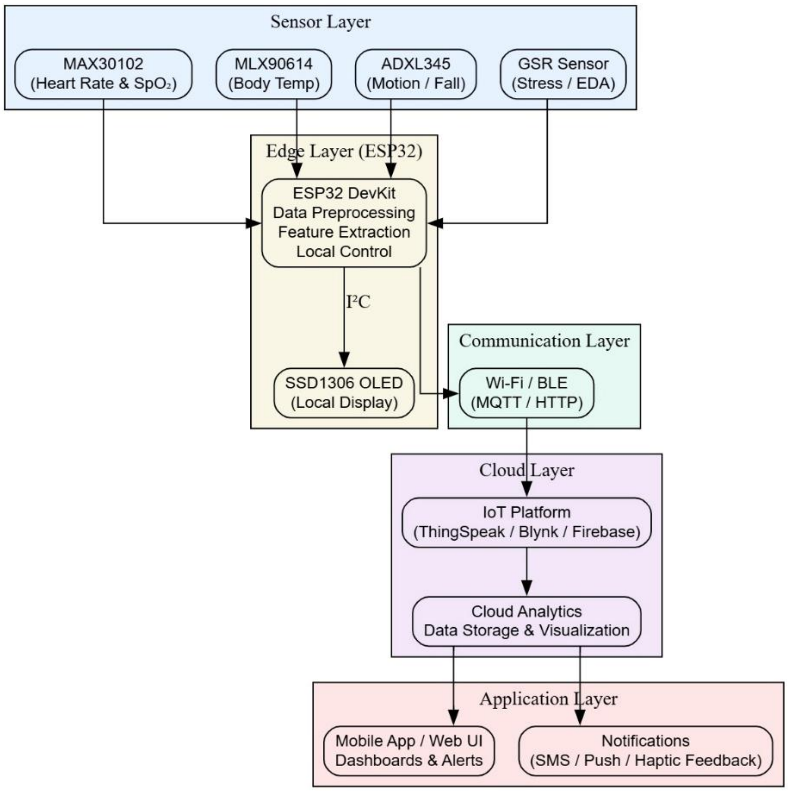

Figure 1.1: Architecture Diagram

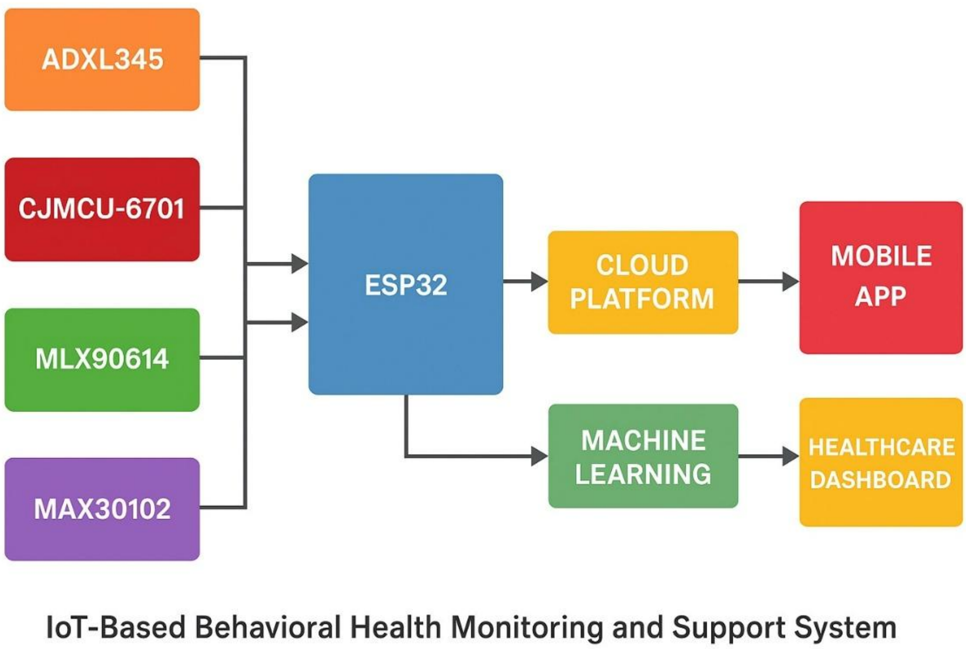

Figure 1.2: Block Diagram

1.2 Relevance of the Project

This report is structured to provide a comprehensive understanding of the proposed IoT-based Behavioral Health Monitoring and Support System. It is organized into the following chapters:

Chapter 1: Introduction – Introduces the background, motivation, objectives, and scope of the project, along with the organization of the report.

Chapter 2: Literature Review – Presents a review of existing systems and research in the area of behavioral health monitoring, highlighting gaps and opportunities.

Chapter 3: System Design and Methodology – Describes the architecture, components, and methodology used for the design and implementation of the system.

Chapter 4: Implementation – Provides details on hardware and software implementation, including the integration of sensors, ESP32 microcontroller, and data processing techniques.

Chapter 5: Results and Discussion – Presents the results obtained from the system, along with analysis, validation, and discussion of its performance.

Chapter 6: Conclusion and Future Work – Summarizes the findings of the project, the benefits of the system, and potential areas for future improvement.

This structured approach ensures a logical flow of information, from conceptualization to implementation and evaluation, making it easier for readers to understand the system and its outcomes.

1. 3. Contribution and Key Features

The proposed IoT-based Behavioral Health Monitoring and Support System contributes to the field of digital healthcare by integrating IoT and AI technologies to enable real-time, data-driven mental health monitoring. It addresses the growing need for continuous and preventive mental health management through intelligent sensing and analysis.

Major Contributions:

IoT and AI Integration: Combines smart sensing and cloud-based analytics to detect stress, anxiety, and mood variations.

Real-Time Monitoring: Provides continuous tracking and instant feedback to users for proactive mental health management.

Remote Access: Offers a professional dashboard that allows healthcare practitioners to monitor users and intervene early.

User-Friendly and Non-Invasive: Ensures comfort and ease of use through wearable sensors and a simple mobile interface.

Cost-Effective Design: Uses affordable components, making the system accessible and scalable for wide applications.

Key Features:

Continuous monitoring of behavioral and physiological parameters.

Cloud-based data processing and machine learning analysis.

Personalized feedback and alerts via a mobile app.

Secure and portable system architecture.

Overall, the project promotes proactive mental health support and demonstrates the potential of IoT in creating intelligent and compassionate healthcare solutions.

1.4 Motivation and Objectives

Motivation

In today's fast-paced lifestyle, stress, anxiety, and emotional imbalance have become common issues affecting mental well-being. Traditional assessment methods, such as clinical visits or self-reports, are often subjective and fail to provide real-time monitoring. The motivation for this project arises from the need for a **continuous and data-driven behavioral health monitoring system** using IoT technology. By integrating sensors, cloud platforms, and analytics, the system can track key parameters like heart rate and temperature to detect behavioral changes early. This approach promotes **proactive mental health management** and supports timely intervention.

Objectives

The main objective of this project is to design and develop an **IoT-based Behavioral Health Monitoring and Support System** that continuously monitors and analyzes behavioral health indicators.
The specific objectives are:

1. To design an IoT-based setup for collecting physiological and behavioral data.
2. To monitor parameters such as heart rate, temperature, and activity levels related to emotional states.
3. To store and visualize real-time data using cloud integration.
4. To analyze behavioral trends and provide personalized feedback.
5. To develop a cost-effective and accessible system for continuous mental health support

Chapter 2

Literature Survey

Chapter 2

Literature Survey

2.1 Introduction

The increasing prevalence of mental health challenges such as stress, anxiety, and depression has become a major public health concern in recent years. With the rise of modern lifestyles characterized by long working hours, high digital exposure, and reduced physical activity, behavioral and emotional disorders are becoming more common, especially among young adults and working professionals. However, early detection and continuous monitoring of mental health remain limited due to the lack of accessible, real-time, and data-driven tools.

Traditional mental health assessment methods rely heavily on self-report questionnaires and periodic clinical consultations. These approaches often fail to provide accurate, consistent, or timely information, as they depend on subjective perception and are influenced by social or emotional biases. Consequently, many behavioral health issues go unnoticed until they progress to severe stages, making early intervention difficult.

Recent technological advancements, especially in the **Internet of Things (IoT)**, **wearable sensors**, and **cloud-based computing**, have opened new opportunities in healthcare. IoT enables seamless connectivity between devices, sensors, and data analytics platforms, allowing continuous and remote health tracking. When applied to behavioral health, IoT facilitates the real-time collection of physiological and behavioral data such as heart rate, sleep quality, activity levels, and skin response—parameters closely associated with emotional states.

Furthermore, the integration of **Artificial Intelligence (AI)** and **Machine Learning (ML)** into IoT ecosystems allows for intelligent data analysis and pattern recognition. These technologies can identify early behavioral changes, predict stress levels, and provide personalized feedback or alerts. Combined with mobile health applications and cloud infrastructure, such systems can support users in managing their mental well-being effectively from anywhere and at any time.

Thus, this chapter provides a comprehensive review of existing literature and related work in the domains of behavioral health monitoring, IoT-based healthcare systems, wearable sensor technologies, data analytics, and cloud-based health support systems. It also identifies the research gaps in current methodologies, emphasizing the need for an integrated, real-time, and personalized **IoT-Based Behavioral Health Monitoring and Support System**.

2.2 Conclusions

The literature review and motivation behind this work emphasize the importance of developing intelligent behavioral health monitoring systems using IoT technologies. While existing healthcare systems primarily address physical health, behavioral and mental well-being often remain underrepresented in digital health innovations.

Through this project, an effort is made to bridge this gap by designing a system that integrates

IoT-based sensing, cloud data processing, and intelligent analytics to monitor and support behavioral health in real time. By achieving the stated objectives, the proposed system aims to contribute meaningfully to the ongoing evolution of smart healthcare. It supports the idea of **proactive, personalized, and preventive care**, ensuring that behavioral health monitoring becomes an integral part of everyday life rather than a reactive clinical measure.

Chapter 3

Proposed Technique

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Proposed Technique

3.1 Project Definition

In modern society, mental health challenges such as stress, anxiety, and depression are becoming increasingly prevalent. Traditional methods for assessing mental health face several limitations: Infrequent evaluations: Mental health assessments are typically done during scheduled clinical visits, which cannot capture real-time fluctuations.

Self-reporting limitations: Users may underreport symptoms or fail to recognize early warning signs, delaying intervention. Lack of personalization: Conventional care often does not provide tailored insights to address individual needs. Limited accessibility: Mental health services are often unavailable or inaccessible, especially in remote areas.

These limitations create a critical gap in early detection, continuous monitoring, and timely intervention. There is a clear need for an IoT-based system capable of continuous behavioral health monitoring and personalized support.

.3.2 Project Statement

The project aims to develop an IoT-based Behavioral Health Monitoring and Support System to bridge the gap in mental health care. The system will continuously monitor key behavioral health parameters, analyze real-time data, and provide personalized insights and alerts to help users maintain mental well-being proactively.

3.3 Project Objective

The main goal of this project is to develop an IoT-based Behavioral Health Monitoring and Support System that continuously tracks mental health parameters and provides personalized support for maintaining well-being. The specific objectives are:

1. **Continuous Monitoring of Mental Health Parameters**: To design a system capable of tracking key indicators such as stress levels, heart rate variability, sleep quality, and physical activity in real-time. Continuous monitoring ensures that fluctuations in mental health are detected promptly, allowing timely interventions.
2. **Data Collection and Analysis**: To collect and process the data generated by sensors to identify patterns, trends, and anomalies in an individual's behavioral health. Analyzing this data helps in understanding the underlying causes of stress or mood changes, enabling more informed decisions for mental wellness.
3. **Proactive Mental Health Support**: To provide users with timely alerts, suggestions, and recommendations based on their behavioral patterns. This proactive approach encourages preventive measures and supports better management of stress, anxiety, and other mental health challenges.
4. **User-Friendly Interface**: To develop a mobile or web application that allows users and healthcare providers to easily visualize health data, track progress, and access actionable insights. The interface will be intuitive, ensuring accessibility for people with varying levels of technical proficiency.
5. **Personalized Care**: To deliver tailored recommendations and strategies based on individual health patterns. Personalized guidance ensures that each user receives support that aligns with their unique mental health needs.
6. **Remote Accessibility and Convenience**: To ensure that the system can be accessed from anywhere, providing continuous support and monitoring regardless of the user's location. This makes mental health support more inclusive and widely available.

3.4 Expected Outcomes:

The expected outcomes of the project include:

Real-time behavioral health monitoring for early detection of stress, anxiety, and depression. Clear visualization of health trends for users and healthcare providers.Enhanced awareness and proactive management of mental health. Accessible and personalized support for improved mental well-being.A scalable system ready for integration with additional sensors or predictive analytics in the future.

3.5 Scope of the Project:

The scope of the project includes:

Designing and implementing an IoT-based monitoring system using ESP32 or similar microcontrollers. Integrating sensors for tracking physiological data such as heart rate, sleep, and activity levels. Developing a mobile/web interface for data visualization and alert notifications. Implementing data analytics to identify behavioral health patterns. Limiting the system to monitoring and support only; it will not replace clinical diagnosis or treatment. Potential expansion for AI-based predictions and multi-user support in future iterations.

Table 3.1 Detailed Comparison between Various Sensor Modules Used in IoT-based Behavioral Health Monitoring System.

Component / Module	Function / Parameter Measured	Communication Type	Approx. Accuracy / Sensitivity	Power Requirement	Application in System
ESP32 DevKit V1	Central microcontroller with Wi-Fi & Bluetooth	UART / I²C / SPI	High (dual-core processor)	3.3V	Data processing, sensor control, and wireless data transmission
MAX30102 Sensor	Heart rate and SpO_2 measurement	I²C	±1% (SpO_2), ±2 BPM (HR)	1.8V–3.3V	Continuous vital monitoring for stress detection
Grove GSR Sensor	Measures skin conductivity (EDA)	Analog	Sensitive to micro-Siemens variations	3.3V–5V	Emotional arousal and stress response tracking
ADXL345 Accelerometer	Detects motion, activity, and posture	I²C / SPI	±2g to ±16g range	3V–5V	Monitors movement and behavioral patterns
OLED Display (SSD1306)	Displays vitals and device status	I²C	High contrast 128×64 pixels	3.3V–5V	User feedback and on-device visualization
TP4056 Charging Module	Battery charging and protection	USB	Efficiency >90%	5V input	Safe Li-ion/Li-Po charging
Li-ion Battery (3.7V, 1000–2200 mAh)	Portable power source	—	Long battery life (6–8 hrs)	3.7V	Powers the wearable device
Velcro Strap / Enclosure	Physical housing and wearability	—	Durable / adjustable	—	Ensures comfort and usability

3.1.2 SpO_2 Sensor :

The **SpO_2 sensor module** is a pulse oximetry and heart-rate monitoring sensor designed for wearable and IoT-based health monitoring systems. It operates using two LEDs — **a red LED (660 nm)** and **an infrared LED (880–940 nm)** — along with a **photodetector** to measure the amount of oxygenated and deoxygenated hemoglobin in the blood. By analyzing the light absorption differences between the red and infrared wavelengths, the sensor calculates **SpO_2 (oxygen saturation)** and **pulse rate** in real time.

Features of SpO_2 Sensor :

Measures SpO_2 and heart rate using optical sensing.

Integrated LEDs, photodetector, and signal processing unit in a single chip.

Low power consumption, suitable for wearable devices.

I^2C interface for easy communication with microcontrollers like Arduino or ESP32.

Onboard ambient light cancellation for accurate readings.

Compact size (CJMCU breakout board) for easy integration in IoT and biomedical systems.

Operating Voltage: 1.8 V to 5 V (typically powered via 3.3 V or 5 V).

Fig 3.1: SpO2 sensor

3.1.3 Accelerometer:

An **accelerometer** is a sensor that measures the **acceleration forces** acting on an object along one or more axes. It detects both **static forces** like gravity and **dynamic forces** caused by motion or vibration, making it ideal for monitoring movement, orientation, and physical activity.In IoT applications, accelerometers are widely used to capture motion-related data, detect falls, or analyze activity patterns. They work by converting mechanical motion into electrical signals, which can then be read by a microcontroller such as the **ESP32** for further processing.The sensor typically communicates via **analog output** or digital interfaces like **I2C or SPI**, making it easy to integrate with IoT boards. Accelerometers are lightweight, low-power, and compact, which suits wearable and embedded applications..

Features of Accelerometer (e.g., ADXL335 / MPU6050) :-

Measurement Axes: 3-axis (X, Y, Z)

Measurement Range: ±2g, ±4g, ±8g, ±16g (depends on model)

Interface: I2C / SPI

Supply Voltage: 3.3V – 5V

Resolution: 10–16 bit (depending on sensor)

Applications: Motion detection, fall detection, orientation sensing, activity monitoring.

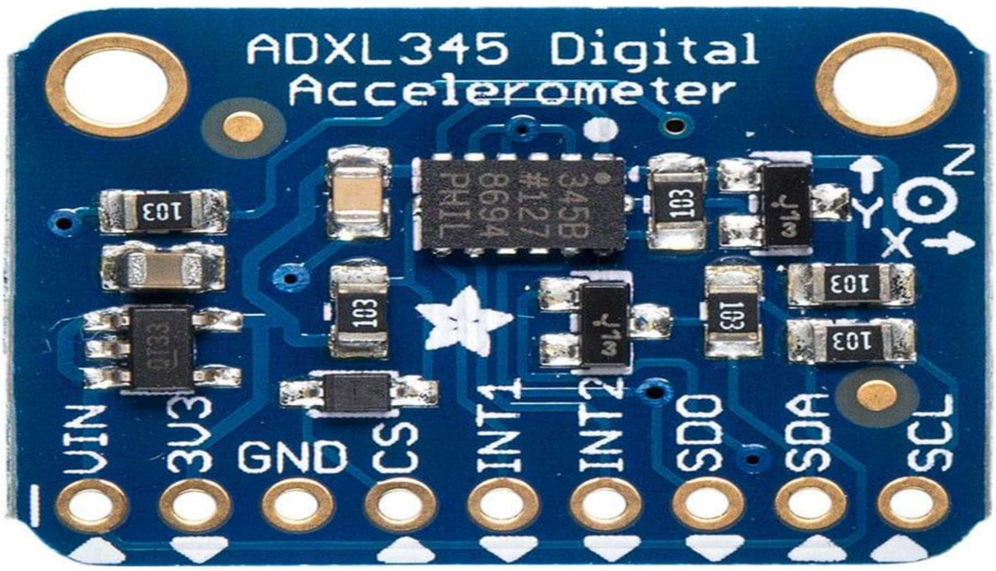


Fig 3.2: Accelerometer

3.1.4 MLX90614 Infrared Temperature Sensor :

The **MLX90614** is a non-contact infrared temperature sensor that measures object and ambient temperatures by detecting emitted infrared radiation. It features a built-in thermopile detector, 16-bit ADC, and I²C interface for accurate digital output. Compact and reliable, it's ideal for human body temperature monitoring and IoT-based healthcare applications.

MLX90614 Infrared Temperature Sensor Specifications & Features :

Non-contact temperature measurement – detects temperature without physical contact, ideal for hygiene-critical and moving objects.

Wide temperature range – measures object temperatures from –70°C to +380°C.

High accuracy and resolution – typically ±0.5°C accuracy with a 0.02°C resolution.

Dual temperature sensing – measures both object and ambient temperatures simultaneously.

Digital I²C interface – easy communication with microcontrollers like Arduino, ESP32, or Raspberry Pi.

Built-in 16-bit ADC and DSP – provides stable, high-precision readings with minimal noise.

Low power consumption – suitable for battery-powered IoT and wearable devices.

Configurable emissivity setting – enables accurate measurement of various surfaces and materials.

Compact module design – easy to integrate into embedded and IoT-based health monitoring systems.

Fig 3.3: MLX90614 Infrared Temperature Sensor

3.1.5. CJMCU GSR (Galvanic Skin Response) :

The CJMCU GSR module detects changes in the electrical conductance of the skin. When a person becomes stressed or emotionally stimulated, sweat gland activity increases, causing the skin's resistance to drop. The module converts these changes into an analog signal that can be read by microcontrollers like **Arduino, ESP32, or Raspberry Pi**.

Features of CJMCU GSR :

- **Measures skin conductivity** to detect emotional or stress responses.
- **Analog output** easily interfaced with ADC pins of microcontrollers.
- **Low power consumption**, ideal for portable or wearable devices.
- **Compact size and easy integration** with IoT boards like ESP32 or Arduino.
- **Onboard signal conditioning circuit** for stable readings.
- **Suitable for behavioral or psychological experiments**, lie detection, or stress tracking.

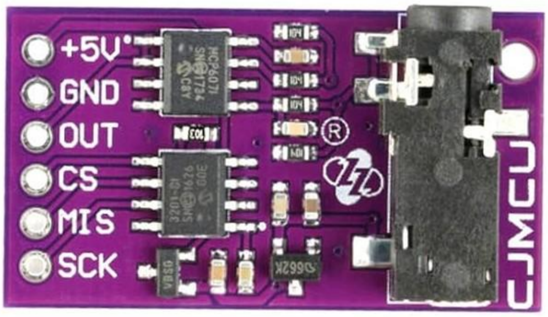


Fig 3.4: Galvanic Skin Response

3.6 Problem Definition

1. **Lack of Continuous & Objective Monitoring:**
   Mental health issues like stress, anxiety, and depression are not tracked continuously. Existing methods depend on self-reporting and occasional clinic visits, which are subjective and often fail to capture real-time physiological or behavioral changes.
2. **Delayed Detection & Ineffective Intervention:**
   Because there is no real-time monitoring system, early symptoms often go unnoticed. This leads to delayed diagnosis, poorly timed interventions, and treatment plans that are not personalized to the patient's daily mental health patterns.
3. **Need for a Non-invasive, User-friendly Monitoring Solution:**
   Traditional assessment methods cannot detect subtle biometric signals related to emotional state. There is a need for a portable, non-invasive system that can automatically collect, analyze, and alert users/caregivers about mental health fluctuations using objective sensor data.

3.7 Working Principle

The working principle of the IoT-Based Behavioral Health Monitoring and Support System shown in the image involves sensor data acquisition, wireless transmission using ESP32, cloud and machine learning integration, and remote health support.

The system uses multiple sensors—**ADXL345 (accelerometer), CJMCU-6701 (gyroscope/accelerometer), MLX90614 (infrared temperature), and MAX30102 (pulse oximeter/heart-rate)**—interfaced with the **ESP32 microcontroller** via **I2C or SPI** protocols. These sensors collect **real-time physiological and behavioral data** such as motion, body temperature, SpO_2, and heart rate. The ESP32 preprocesses or directly transmits this data to a **cloud platform** over Wi-Fi,enabling **continuous monitoring, centralized storage, and visualization** through mobile or web apps. **Machine learning algorithms** analyze the data for predictive insights, including anomaly detection, behavioral classification, and trend analysis.

Key Steps:

1. **Sensor Acquisition:** Sensors measure health and behavioral parameters and send raw data to ESP32.
2. **Microcontroller Processing:** ESP32 performs initial processing and transmits data to the cloud.
3. **Cloud Connectivity:** Data is stored, visualized, and alerts are triggered for abnormal readings.
4. **Machine Learning Integration:** ML models detect risks, trends, or behavioral changes for predictive support.
5. **Remote Access and Support:** Users, caregivers, and doctors access health trends via dashboards and receive timely notifications for intervention.

This architecture allows **real-time monitoring, early intervention, and enhanced patient safety** through automated alerts and data-driven insights.

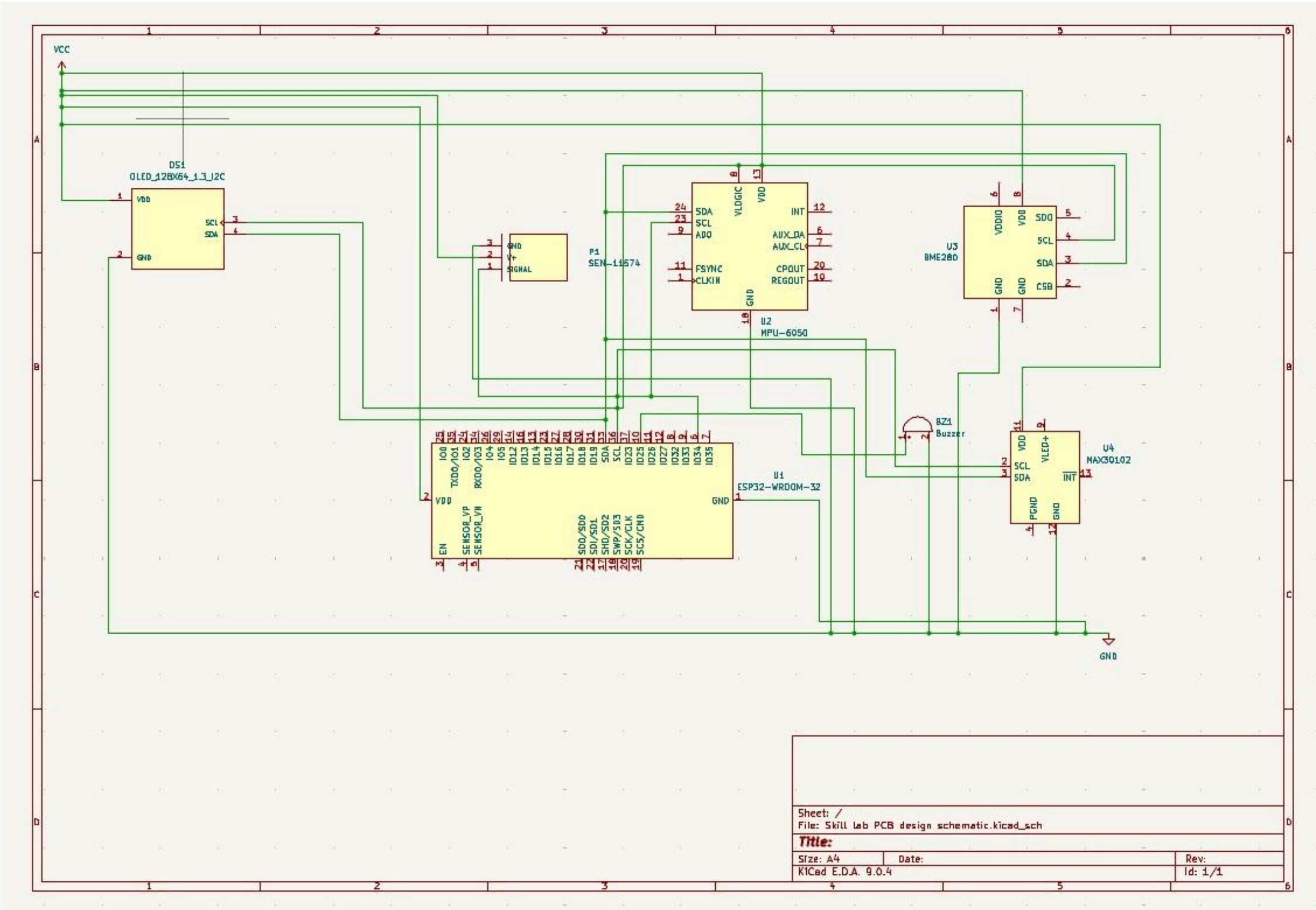


Fig 3.5: Schematic diagram

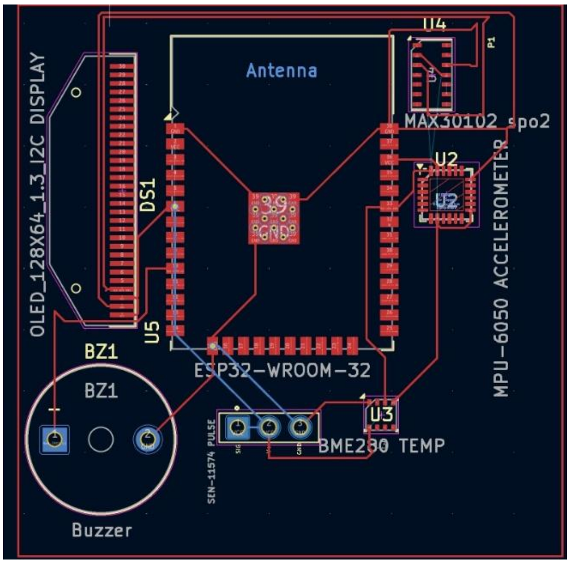


Fig 3.6: PCB Designing

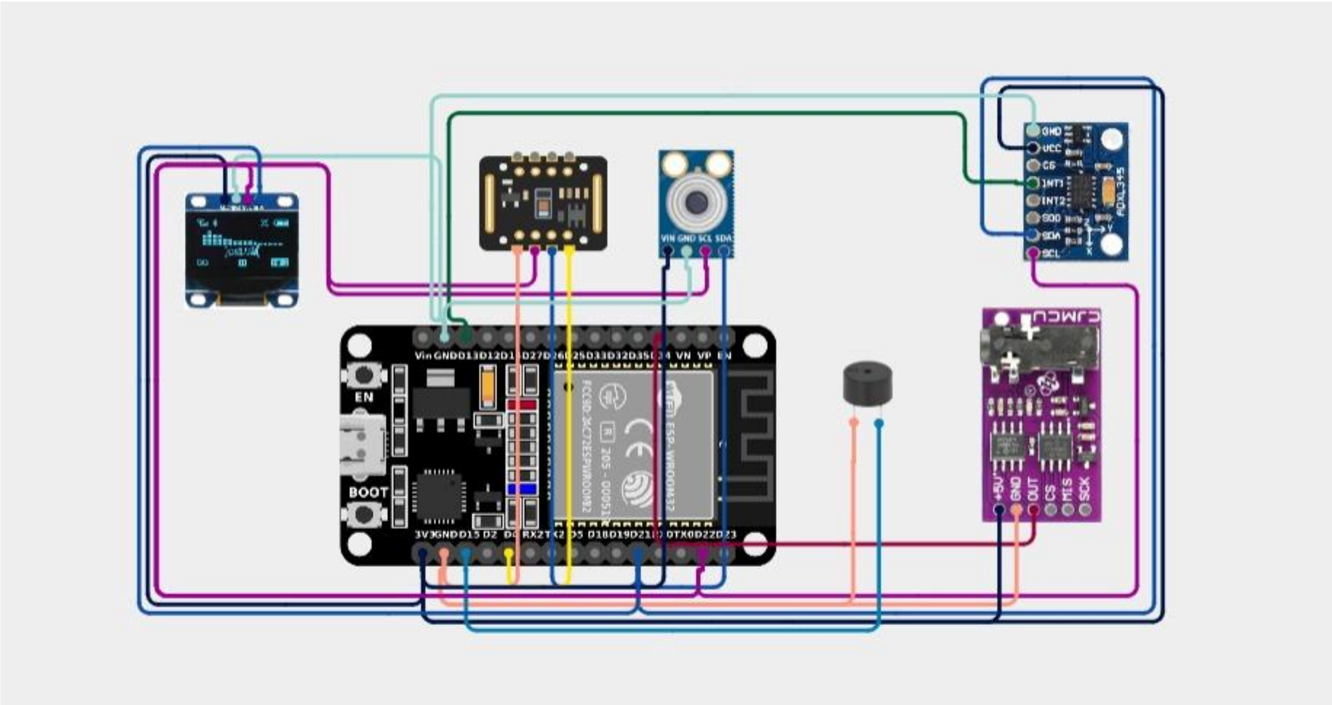


Fig 3.7: Circuit Diagram

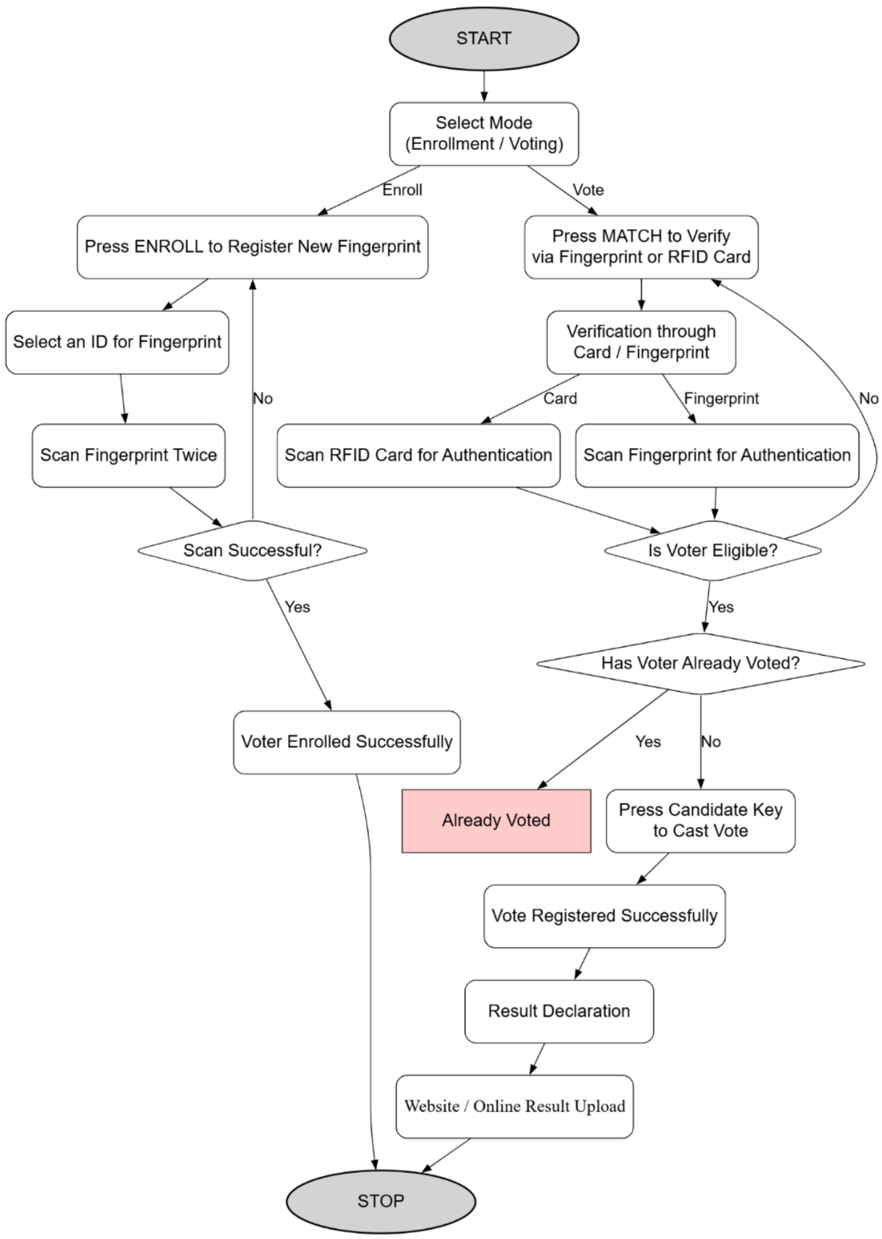


Fig 3.8: Flow chart for proposed model

The flowchart describes the working of a proposed electronic voting system that operates in two modes: enrollment and voting. In the enrollment mode, a new voter registers by selecting an ID and scanning their fingerprint twice. If the fingerprint scan is successful, the system confirms that the voter has been enrolled. In the voting mode, the voter verifies their identity either through an RFID card or fingerprint. The system checks whether the voter is eligible and ensures they have not already voted. If the voter has not voted before, they can select their preferred candidate, and the vote is recorded successfully. After voting, the system proceeds to result declaration and online result upload before ending the process.

Chapter 4

Result Analysis

4.1 Result Analysis

1. GSR Sensor (CJMCU670)

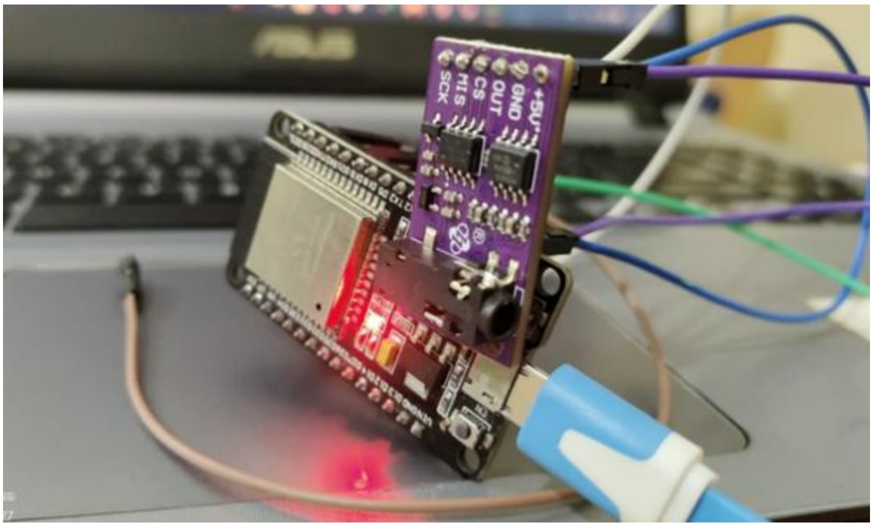
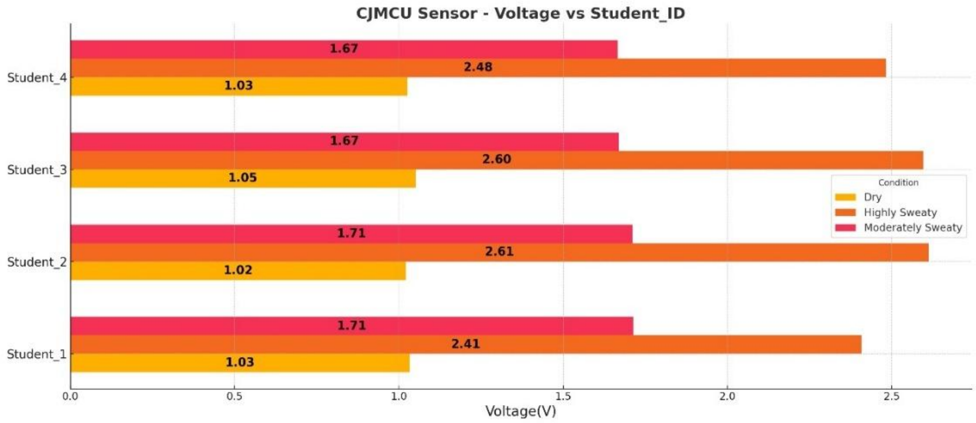
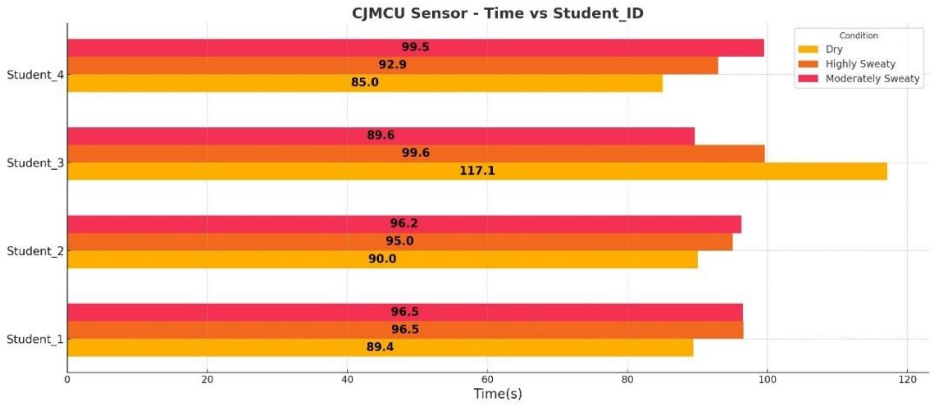


Fig 4.1: CJMCU GSR

Table 4.1.1 Readings of GSR Sensor (CJMCU670)

Student_ID	Condition	Time(s)	Voltage(V)
Student_1	Dry	89.41176	1.032294
Student_1	Highly Sweaty	96.53846	2.406923
Student_1	Moderately Sweaty	96.48649	1.713189
Student_2	Dry	90	1.0205
Student_2	Highly Sweaty	95	2.6123
Student_2	Moderately Sweaty	96.25	1.711583
Student_3	Dry	117.1429	1.05
Student_3	Highly Sweaty	99.58333	2.595375
Student_3	Moderately Sweaty	89.59184	1.668816
Student_4	Dry	85	1.025571
Student_4	Highly Sweaty	92.91667	2.482167
Student_4	Moderately Sweaty	99.52381	1.6655





2. Temperature Sensor (MLX90614)

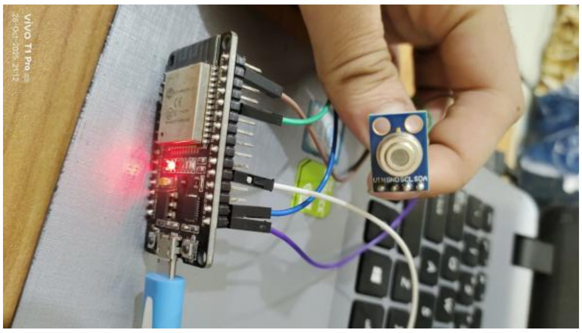
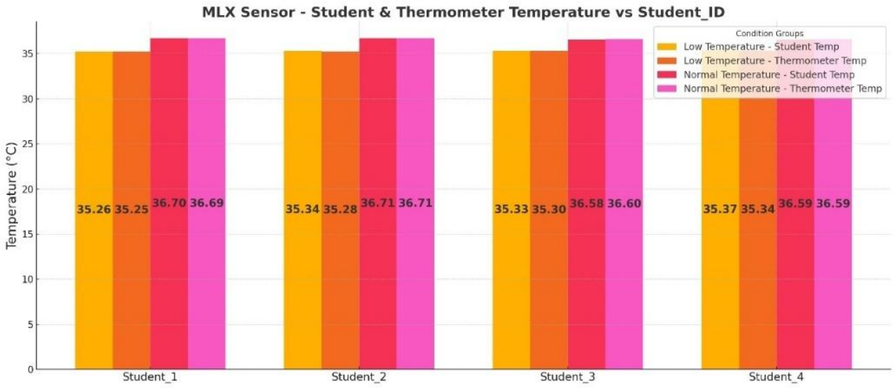


Fig 4.2: Temperature Sensor

Table 4.1.2 Readings of Temperature Sensor (MLX90614)

Student_ID	Condition	Time(s)	RoomTemp(°C)	Student(°C)	Thermometer(°C)
Student_1	Low Temperature	88.15789	28.865	35.25947	35.25079
Student_1	Normal Temperature	101.1905	30.09524	36.70048	36.69048
Student_2	Low Temperature	94.0625	28.84656	35.33531	35.27563
Student_2	Normal Temperature	95.625	30.15542	36.71125	36.70563
Student_3	Low Temperature	95.42857	28.95171	35.32829	35.296
Student_3	Normal Temperature	94.66667	30.05667	36.57733	36.60022
Student_4	Low Temperature	95.13514	28.90784	35.36541	35.33757
Student_4	Normal Temperature	94.88372	30.10349	36.58581	36.59093



3. SpO2 (MAX30102)

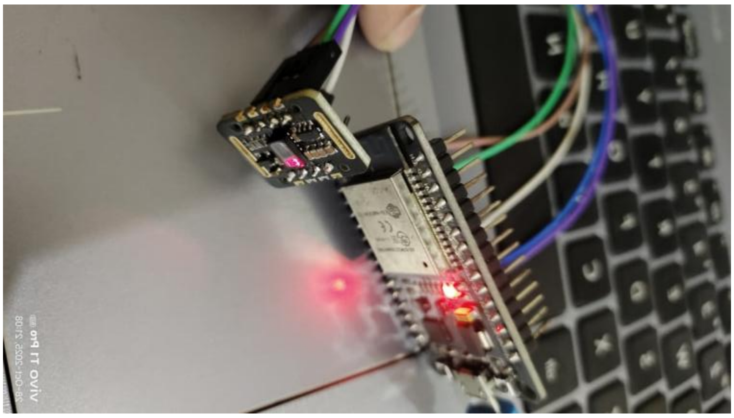


Fig 4.3: SpO2

Table 4.1.3 Readings of SpO2 (MAX30102)

Student_ID	Condition	BPM	SpO2(%)
Student_1	Active	116.1545	96.82273
Student_1	Normal Activity	83.50732	97.72927
Student_1	Post Exercise	123.725	97.05
Student_1	Resting	63.79231	98.2
Student_2	Active	113.6409	96.91818
Student_2	Normal Activity	82.31892	97.82703
Student_2	Post Exercise	121.0833	97.33333
Student_2	Resting	65.34667	98.07333
Student_3	Active	117.2238	96.87143
Student_3	Normal Activity	83.15897	97.76667
Student_3	Post Exercise	131.4	96.85
Student_3	Resting	64.9	98.20556
Student_4	Active	118.9095	96.76667
Student_4	Normal Activity	83.98889	97.80667
Student_4	Post Exercise	116.925	96.9
Student_4	Resting	63.47	98.14

4. **Accelerometer**

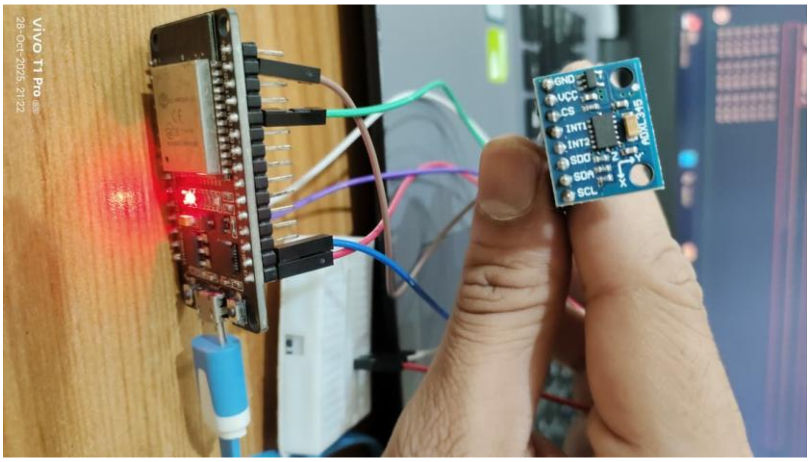


Fig 4.4: Accelerometer

4.2 Result Verification

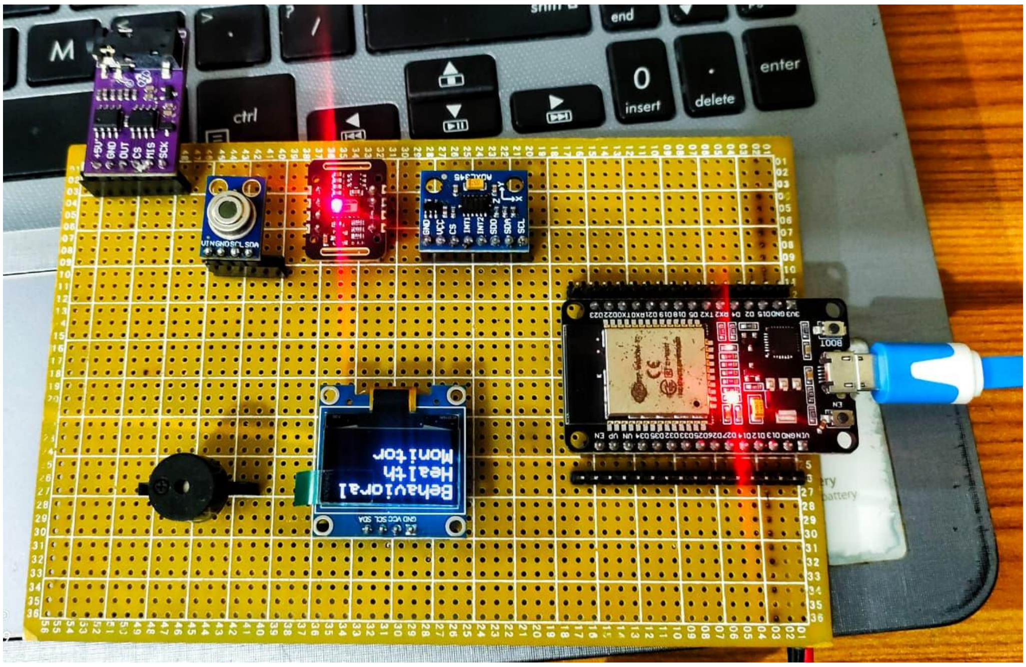


Fig 4.5: Integrated Circuit

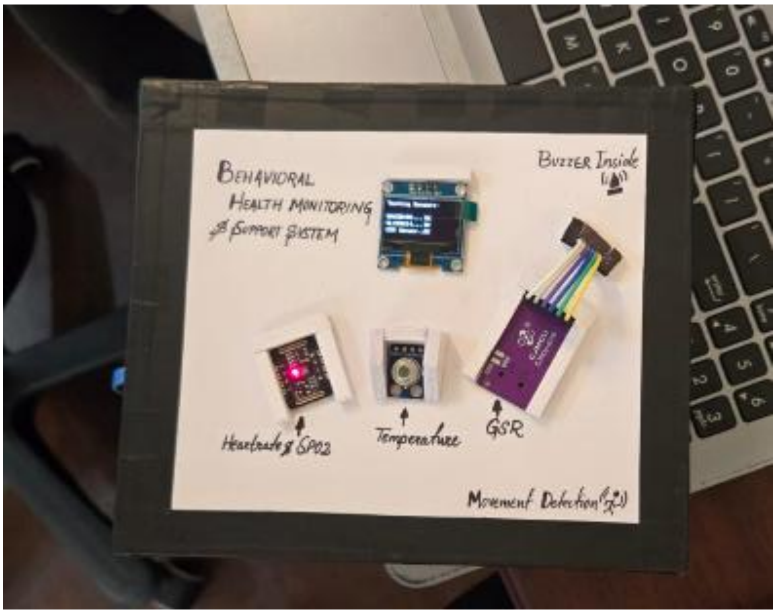


Fig 4.6: Top view of our device

Fig 4.7: Side view of our device

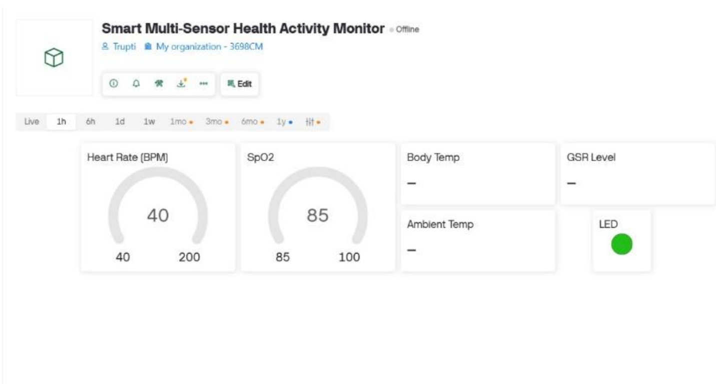


Fig 4.8: Dashboard

This proposed model was designed to serve the purpose of continuously monitoring an individual's behavioral and physiological parameters using IoT technology, with the goal of detecting stress, emotional fluctuations, and early mental health variations. As stated in the earlier chapters, the system integrates multiple biometric and motion-based sensors such as MAX30102, GSR sensor, and ADXL345 along with the ESP32 microcontroller. Although mental health cannot be fully diagnosed through sensors alone, this model significantly enhances early detection by identifying abnormal patterns, reducing human error, and enabling real-time intervention through continuous and objective monitoring.

The operations carried out in this proposed IoT-based behavioral monitoring system include collection of physiological data through sensors, preprocessing and filtering of data by the ESP32 board, transmission of processed data to a cloud platform, analysis and pattern detection to identify stress levels, alert generation when abnormal patterns occur, and display of real-time values on a user interface so that both users and professionals can view the health status remotely.

In this proposed system, when the user wears the device the sensors immediately start collecting biometric data.

The MAX30102 sensor reads heart rate, HRV, and SpO_2 values, while the GSR sensor measures skin conductance to determine emotional arousal. The ADXL345 accelerometer continuously tracks movement and activity level. All these readings are sent to the ESP32 for analysis.

Once the sensors capture the data, the ESP32 processes it by filtering noise and identifying unusual patterns.

If the heart rate suddenly increases, GSR values rise significantly, or if activity level drops unexpectedly, these changes are recognized by the controller. After preprocessing, the ESP32 transmits the data to the cloud server through Wi-Fi so that it can be accessed and monitored remotely.

The cloud platform plays a crucial role in analyzing the user's physiological and behavioral trends.

After receiving the uploaded data, the system compares it with predefined thresholds. If the data indicates stress, anxiety, or abnormal behavior, the cloud triggers an alert. These alerts are displayed instantly on the user dashboard or app so that the user can take immediate action.

The system ensures reliability and avoids false readings by not allowing unverified or corrupted values to be processed.

If the data captured by sensors is incomplete or noisy, the ESP32 discards it and prompts a fresh reading. This ensures that the user receives accurate information and that no wrong analysis is performed due to faulty sensor data.

During continuous monitoring, if the system detects a stress episode, it instantly notifies the user.

The alert message may include suggestions such as taking a break, trying relaxation techniques, or contacting a professional. The system stores all the alerts and readings on the cloud so that they can be reviewed later by doctors, therapists, or caregivers.

At the end of a monitoring session, the dashboard automatically updates the summary of the user's health condition.

The system displays graphs of heart rate, GSR fluctuations, activity patterns, and stress episodes. This is followed by storing the daily report on the cloud platform, allowing users and professionals to access the data anytime from anywhere, provided they have login credentials.

Chapter 5

Conclusions

Chapter 5

5.1 Conclusions

The development of an IoT-Based Behavioral Health Monitoring and Support System represents a significant step toward integrating technology with mental health management. Throughout this project, an effort was made to design a system capable of continuously monitoring key physiological parameters such as heart rate, body temperature, and physical activity to infer behavioral health conditions.

By combining IoT hardware (sensors and ESP32 microcontroller) with cloud-based data storage and analytics, the proposed system demonstrates how technology can help individuals track their emotional well-being in real time. The data collected can serve as a reliable indicator of behavioral patterns and assist users in understanding their stress levels, fatigue, or emotional fluctuations more accurately.

This project highlights the importance of continuous, objective, and data-driven monitoring in mental health, overcoming the limitations of traditional self-assessment methods. The results of this work emphasize that IoT can effectively bridge the gap between physical and mental healthcare, promoting a more holistic approach to personal wellness.

Overall, the proposed system provides a foundation for future research and development in digital behavioral healthcare, supporting the goal of proactive and preventive mental health management rather than reactive treatment.

5.1 Scope for future investigations

While this project successfully demonstrates the concept and functionality of IoT-based behavioral health monitoring, there are several areas where future enhancements can be explored to improve system accuracy, reliability, and user experience:

1. **Integration of Advanced Sensors:**
   Future versions of the system can incorporate additional biosensors such as ECG, EEG, and GSR sensors for deeper insight into emotional and stress-related responses.
2. **Machine Learning and AI Algorithms:**
   Implementing advanced data analytics and AI models can help predict behavioral trends, classify stress levels, and generate personalized recommendations automatically.
3. **Mobile Application Development:**
   A dedicated mobile app can be developed to provide real-time visualizations, instant alerts, and user-friendly dashboards for behavioral insights.

4. **Cloud and Data Security Enhancements:**
   Improved encryption and secure authentication methods can ensure privacy and protect sensitive behavioral data stored on cloud platforms.
5. **Clinical Validation and Testing:**
   The system can be tested and validated with healthcare professionals to ensure medical accuracy and real-world applicability for stress and mood detection.
6. **Integration with Smart Environments:**
   Future systems could be connected with smart home or wearable ecosystems, enabling automatic adjustments in lighting, sound, or temperature based on the user's emotional state.

By implementing these improvements, the system can evolve into a **comprehensive behavioral health management platform**, capable of offering personalized, intelligent, and clinically reliable mental health support to users.

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